Aerial Sentinel AI: Bird & Drone Detection System

Subtitle: *Intelligent Computer Vision for Airspace Monitoring*

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Introduction & Objective

The Problem: Increasing drone activity creates security risks; difficult to distinguish small drones from birds at a distance.

The Goal: Build an automated detection system to classify and localize aerial objects in real-time.

Target Users: Security personnel, airport authorities, and wildlife researchers.

Technical Architecture

Core Engine: YOLOv8s (You Only Look Once - Version 8 Small).

Why YOLOv8?: High accuracy on small objects with extremely low latency.

Frontend: Streamlit Web Framework.

Cloud Infrastructure: Streamlit Cloud (Linux/Debian).

Dataset & Training Process

lasses: Two-class classification (Bird, Drone).

Data Prep: Images resized to 640×640 pixels for optimal feature extraction.

Training Specs: * **Epochs:** Fine-tuned for stability.

GPU: Trained on NVIDIA GeForce GTX 1650 Ti.

Optimizer: SGD/Adam with manual FP32 precision.

Overcoming Hardware Challenges

The AMP Issue: Encountered "Automatic Mixed Precision" warnings on GTX 1650 Ti hardware.

Risk: Potential for NaN (Not a Number) losses and model collapse.

Strategy: Disabled AMP to prioritize numerical stability over training speed.

Result: A robust weight file (best.pt) with no loss anomalies.

System Performance & Metrics

mAP50 (Mean Average Precision): Achieved **0.7654**.

Inference Speed: Average **3.9ms to 10.2ms** per frame.

Recall: High recall to ensure no drones are missed in critical airspace.

Efficiency: Lightweight model architecture suitable for edge deployment.

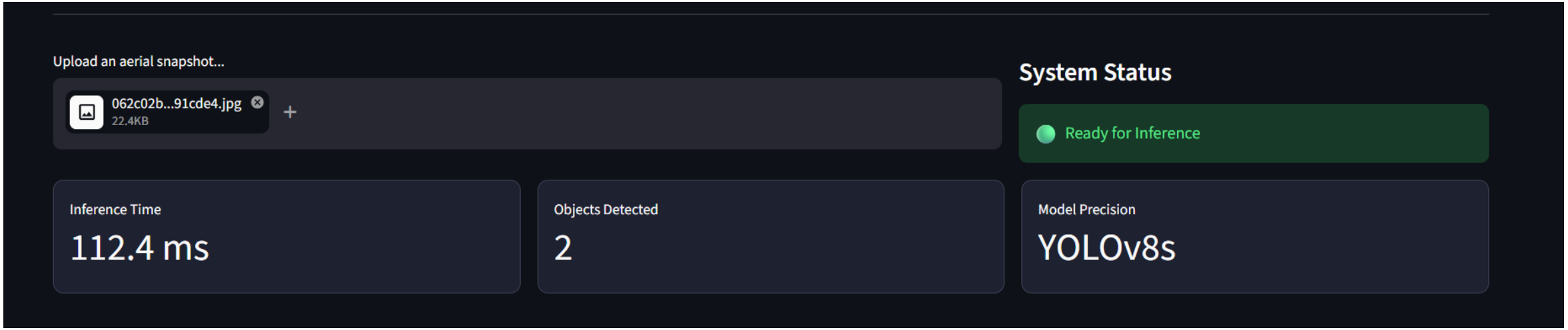
Deployment & DevOps

Linux Environment: Solved dependency issues with libGL.so.1 and libgthread-2.0.so.0.

CI/CD Pipeline: Automated deployment via GitHub.

Resource Management: Implemented @st.cache_resource to keep the AI model "warm" in server memory for instant responses.

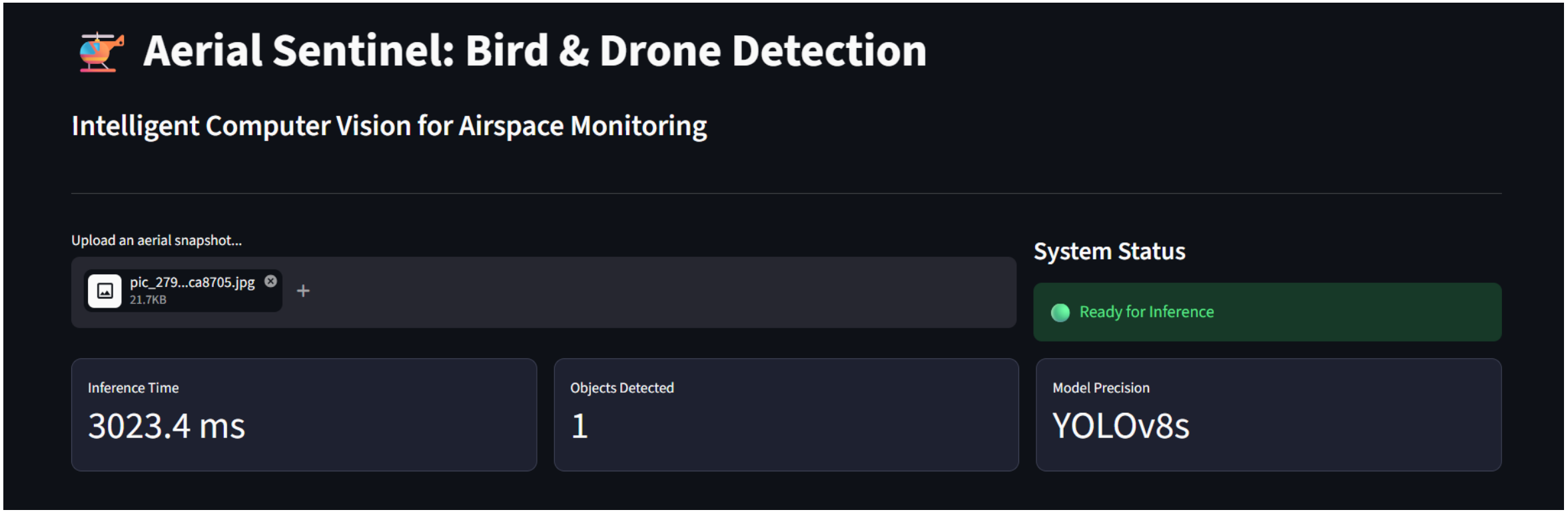
OUTOUT IMAGE BIRD



View Detection Meta-Data

```
tensor([[1.1259e+01, 7.9598e+01, 2.1776e+02, 3.9791e+02, 5.8539e-01, 0.0000e+00],
        [3.4320e+02, 1.5763e+02, 4.1600e+02, 3.2793e+02, 2.7725e-01, 1.0000e+00]])
```

OUTOUT IMAGE DRONE



Original Image

AI Visual Output

View Detection Meta-Data

```
tensor([[128.7056,  75.2173, 381.0576, 199.8350,   0.8450,   1.0000]])
```

Dashboard Features (Demo)

Interactive UI: Dark-mode command center layout.

Real-time Analytics: Displays inference latency and object count metrics.

Adjustable Thresholds: User-controlled slider to filter sensitivity based on environmental conditions.

Challenges & Key Learnings

1

Challenge: Migrating from Windows (Local) to Linux (Cloud) environment.

2

Challenge: Handling small object detection in cluttered backgrounds.

3

Key Learning: Mastered the full AI lifecycle: Data -> Training -> Optimization -> Cloud Deployment.

Conclusion & Future Work

Conclusion: Successfully delivered a production-ready AI tool that meets Labmentix performance standards.

Future Scope: * Integrate RTSP (Real-Time Streaming Protocol) for live CCTV feeds. Expanding classes to include airplanes and helicopters.

THANK YOU